When AI Is the Target: Real-World Adversarial Attacks and Defenses

One threat, adversarial manipulation of AI systems, traced through three real cases, three company defenses and the rules now taking shape

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THREAT AND IMPACT

The threat: tricking AI into the wrong decision

Evasion

Altered input fools a trained model (a sign, an image)

Poisoning

Corrupted training data or model components

Prompt injection

Hidden instructions hijack an AI assistant

Deepfakes

AI-faked faces and voices fool people and checks

US$25 million

lost in one deepfake video call, Hong Kong, January 2024

US$40 billion

projected US fraud losses enabled by generative AI by 2027 (32% a year)

40+ million

vehicles use the camera that a strip of tape fooled

Most exposed industries: autonomous vehicles (safety), finance (money moved on a faked instruction) and healthcare (life-critical model decisions).

Sources: Lalchand et al. (2024); Povolny & Trivedi (2020); Parisi (2019, Ch. 10).

STEP 1 · CASE STUDIES

Three real attacks that made headlines

AUTONOMOUS VEHICLES · EVASION

Tesla: tape on a speed sign

How: 5 cm of black tape on a 35 mph sign made the MobilEye camera read 85.

Impact: The car began accelerating toward 85 mph; the camera is in 40+ million vehicles.

Outcome: Newer cameras handle it; older cars were not updated.

FINANCE · DEEPFAKE

Hong Kong: a fake CFO on video

How: An employee joined a video call where the CFO and colleagues were all deepfakes.

Impact: US$25 million sent to the fraudsters.

Outcome: Money lost; the firm and its staff were tricked, not its IT systems.

ENTERPRISE AI · PROMPT INJECTION

EchoLeak: Microsoft 365 Copilot

How: A crafted email could make Copilot leak data with zero clicks.

Impact: Rated **9.3 Critical** (CVSS).

Outcome: Microsoft fixed it server-side; no exploitation was reported.

WHY THEY MADE HEADLINES

None needed a break-in: each fooled a trusted AI, or a person, with input crafted for it.

Sources: Povolny & Trivedi (2020); Lalchand et al. (2024); Microsoft Security Response Center (2025).

STEP 2 · CORPORATE DEFENSES

How Microsoft, Google and Meta defend their AI

MICROSOFT · RED TEAMING

Attack it before attackers do

Red-teamed **100+** generative AI products.

Lessons: attacks need no gradients, humans stay crucial, security is never finished.

GOOGLE · ADVERSARIAL TRAINING

Layered defense for Gemini

Training on adversarial data “significantly enhanced” resistance to prompt injection.

Plus ML classifiers, link redaction and user confirmation.

META · OPEN GUARDRAILS

LlamaFirewall for AI agents

PromptGuard 2 detects jailbreaks; **AlignmentCheck** audits agent reasoning.

CodeShield blocks insecure generated code.

WHAT IS PROVING MOST EFFECTIVE

Layers, not one fix: adversarial training + input/output classifiers + continuous red teaming + a human check before money moves.

Sources: Bullwinkel et al. (2025); Google GenAI Security Team (2025); Chennabasappa et al. (2025).

COMPARISON CHART

Industry comparison: attack types vs. defense strategies

Attack type	Industry most hit	Adversarial training	Red teaming & testing	Input/output classifiers	Anomaly detection	Human verification
Evasion	Autonomous vehicles	●	●	◐	●	—
Poisoning	Healthcare, finance	◐	●	—	●	◐
Prompt injection	Enterprise AI assistants	●	●	●	◐	●
Deepfakes	Finance	—	◐	◐	●	●

● strong fit ◐ partial fit — little effect Author's assessment from the cases and company defenses on slides 3–4.

Takeaway: only red teaming and anomaly detection help against every attack type; deepfakes are stopped by process (call-back checks), not by retraining.

Sources: Google GenAI Security Team (2025); Bullwinkel et al. (2025); Chennabasappa et al. (2025); Parisi (2019, Ch. 5).

Applying Chapter 5, and the governance trend that requires it

CHAPTER 5: ANOMALY DETECTION AS A DEFENSE

Model what normal looks like, then flag what does not fit.

- **Inputs:** a camera reading of 85 mph on a 35 mph road is out of distribution → slow down, ask the driver.
- **Payments:** a large, urgent transfer right after one video call is unusual → hold and call back.
- **Training data:** outliers in new data are screened before retraining → stops poisoning.
- **AI usage:** an assistant suddenly sending data to an outside link → block it.

TREND: AI GOVERNANCE MAKES ROBUSTNESS LAW

The **EU AI Act, Article 15**, requires high-risk AI to be "resilient against attempts by unauthorised third parties to alter their use, outputs or performance."

It names data poisoning, model poisoning and adversarial examples, and asks for measures to "prevent, detect, respond to" them.

Why it matters: robustness testing and anomaly detection move from best practice to legal duty.

Sources: Parisi (2019, Ch. 5); Regulation (EU) 2024/1689 (2024, Art. 15).

Reflection and references

Reflection. In Modules 3–5 I built detectors and trusted their accuracy; this module showed that the detector itself is a target. I want to build AI defenses that are red-teamed before release and backed by anomaly checks.

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Chapter citations: Parisi (2019), Chapter 5 “Network Anomaly Detection with AI” and Chapter 10 “Assessing your AI Arsenal.”